HAWASSSA UNIVERSITY

MOBILE APPLICATION DEVELOPMENT

GROUP PROJECT

Med-Tracker ET

Project Proposal — Phase 1

Your digital bridge to better medication management

1. Project Overview

Med-Tracker ET is a mobile Android application designed to address a critical healthcare gap in Ethiopia. The application serves as a digital bridge connecting patients with their medications by providing real-time pharmacy stock information, personalized medication reminders, prescription management, and a comprehensive drug information guide — all accessible from a standard Android smartphone.

This project is developed as part of the Mobile Application Development course at Hawassa University. The app targets the everyday Ethiopian patient who struggles with medication adherence and pharmacy accessibility, two of the most pressing healthcare challenges in the country today.

2. Problem Statement

Ethiopia's healthcare system faces significant structural challenges that directly impact patients' ability to manage their medications effectively. Two major issues stand out:

> **Drug Stockouts:** Pharmacies across Ethiopia frequently run out of essential medications, forcing patients to visit multiple pharmacies before finding their prescription. This results in delayed treatment, increased costs, and in severe cases, life-threatening gaps in medication schedules.
>
> **Irrational Drug Use:** Many patients, particularly those managing chronic conditions, struggle with taking the correct dose at the correct time. Without reminders or structured tracking tools, patients frequently miss doses, double-dose by mistake, or stop treatment prematurely — all of which worsen health outcomes and contribute to drug resistance.

These two problems are compounded by a lack of accessible health information in plain language. Most patients do not fully understand their medications, leading to misuse and preventable complications. Med-Tracker ET addresses all three dimensions of this problem through a single, unified platform.

3. Target Users & User Persona

Med-Tracker ET is designed for Ethiopian patients who manage ongoing medication schedules and face challenges with pharmacy access or adherence. The primary target users include:

- Patients managing chronic conditions such as diabetes, hypertension, or tuberculosis
- Caregivers managing medications on behalf of elderly family members
- Urban and semi-urban residents with access to an Android smartphone

Primary User Persona

Attribute	Details
Name	Dawit Tesfaye
Age	45 years old
Location	Addis Ababa, Ethiopia
Occupation	Small business owner
Condition	Managing hypertension requiring daily medication
Device	Android smartphone (mid-range)
Pain Points	Visits 2-3 pharmacies before finding medication in stock; misses doses due to a busy sched
Goal	Easily find medications nearby and never miss a dose again

4. Proposed Solution

Med-Tracker ET provides a unified Android platform that solves the medication management problem from multiple angles simultaneously. Rather than requiring patients to rely on memory, paper prescriptions, or multiple pharmacy visits, the app centralizes everything into one simple interface.

The app allows patients to search for nearby pharmacies by medication name and view real-time stock availability, set up personalized daily reminders with dosage instructions for each of their medications, maintain a digital log of their prescriptions including refill dates, and access clear drug information written in plain English. Together, these features significantly reduce missed doses, wasted pharmacy trips, and medication errors.

5. Core MVP Features

The following four features represent the Minimum Viable Product (MVP) scope for Phase 1. Each feature directly addresses one or more of the identified user pain points.

#	Feature	Description	Pain Point Addressed
1	**Medication Reminder & Scheduler**	Users add medications with dosage details and set daily reminder notifications. The app alerts them at the right time each day.	Missed doses, incorrect dosing
2	**Pharmacy Stock Locator**	Patients search by medication name and see a list of nearby pharmacies with real-time stock availability and distance.	Wasted pharmacy trips, stockouts
3	**Prescription Tracker**	A digital log of all active prescriptions showing drug name, dosage, start date, refill date, and treatment progress.	Disorganized prescription management
4	**Medication Information Guide**	A searchable database of common medications with plain-language descriptions of usage, dosage, and side effects.	Irrational drug use, misinformation

6. Technical Approach

Med-Tracker ET will be built as a native Android application using the following technologies and tools:

Component	Technology / Tool
Programming Language	Kotlin
UI Framework	Android Jetpack (XML Layouts)
Local Database	Room (SQLite)
Notifications	Android AlarmManager & NotificationManager
Maps & Location	Google Maps API
Version Control	Git & GitHub
Project Management	Jira (Scrum)
UI/UX Design	Figma

7. Team Members & Roles

The project team consists of five members, each assigned a specific technical role to ensure clear ownership and accountability across all aspects of development.

Name	Role	Responsibilities
FITSUM DAWIT	Core Logic Lead	Business logic, feature implementation, app architecture
KALEAB SHEWADEG	Network & API Lead	API integration, data fetching, pharmacy stock connectivity
FITSUM MOLLA	UI/UX Lead	Figma wireframes, screen layouts, user experience design
BINIYAM GELEBO	Database & State Lead	Room database setup, data models, state management
(Member 5)	DevOps & Testing Lead	GitHub setup, CI/CD, testing, final submission

8. Project Plan — Sprint 1

Sprint 1 is titled 'Project Setup & Design' and covers the full Phase 1 deliverables. Duration: 1 week.

Task	Assigned To	Status
Set up GitHub repository with Android .gitignore	DevOps & Testing Lead	To Do
Create Figma wireframes for core screens	UI/UX Lead	To Do
Write and finalize Project Proposal PDF	Core Logic Lead	To Do
Set up Jira board with Epics and User Stories	Database & State Lead	To Do
Research pharmacy API integration options	Network & API Lead	To Do

9. Expected Outcomes

Upon successful completion of Phase 1, the project will have produced the following deliverables:

- A fully documented project proposal outlining the problem, solution, and team structure
- A GitHub repository with Android .gitignore, README with project links, and /docs folder
- Low-fidelity Figma wireframes for all 5 core screens with a connected user flow
- A configured Jira Scrum board with 4 Epics, 4 User Stories, and an active Sprint 1